CHAN YI HENG

+65 8862 0913 | c@chanyh.com | chanyh.com | cyh-chan | chanyiheng |

Education

National University of Singapore Singapore, SG
Master of Science in Statistics *Aug 2026 – Expected 2027*

Singapore University of Social Sciences Singapore, SG
Bachelor of Science (Honours) in Mathematics *Jan 2022 – Dec 2025*

Singapore Polytechnic Singapore, SG
Diploma in Banking & Finance *Apr 2016 – Mar 2019*

Experience

Teaching Assistant Jan 2026 – Apr 2026
Singapore University of Social Sciences *Singapore, SG*

- Facilitated technical tutorials in Statistical Analysis covering probability theory, asymptotic inference, and classical linear regression.
- Formulated structural review frameworks to ground undergraduate cohorts in deductive mathematical reasoning and problem-solving.
- Guided student execution of formal model diagnostics, emphasizing verification of Gauss-Markov assumptions and residual structures.

Research Assistant Oct 2025 – Present
Singapore University of Social Sciences, University Research Office *Singapore, SG*

- Formulated statistical inference frameworks for experimental psychometric data to quantify treatment effects on latent creativity constructs.
- Developing stochastic models for financial time-series data to capture non-stationarity, temporal dependencies, and volatility dynamics.
- Applying survival analysis and classification techniques to clinical datasets to isolate features driving patient non-attendance.
- Engineered reproducible data pipelines for high-dimensional institutional datasets to ensure mathematical integrity for academic review.

Research Intern Nov 2024 – Aug 2025
*A*STAR, Advanced Remanufacturing and Technology Centre* *Singapore, SG*

- Architected deep learning models for multi-horizon time-series forecasting under non-stationary demand conditions.
- Implemented a SHAP-based Explainable AI (XAI) framework to quantify localized feature attributions and map non-linear interactions.
- Evaluated contemporary deep learning baselines, optimizing hyperparameter spaces to enhance empirical robustness and performance bounds.

Finance Operations Analyst Sep 2021 – Dec 2021
Indeed *Singapore, SG*

- Audited cross-border financial data systems to ensure high-fidelity data migration and compliance with international reporting frameworks.
- Optimized workflows within Salesforce environments by implementing data validation protocols to minimize transactional errors.

Finance Specialist Aug 2019 – Dec 2021
Singapore Armed Forces *Singapore, SG*

- Managed operational budgets and resource allocations under central government regulatory frameworks.
- Executed internal audits and risk assessments, applying structured methodologies to report compliance metrics to leadership.
- Administered large-scale operational budgets during COVID-19 deployments, maintaining accuracy under highly volatile conditions.

Wealth Management Intern Sep 2018 – May 2019
DBS Bank *Singapore, SG*

- Conducted quantitative operational analyses, leveraging process data to isolate system inefficiencies and formulate optimization strategies.
- Enforced compliance protocols through the systematic validation of Accredited Investor documentation and data control frameworks.

Research

Predictive Analytics & Financial Time-Series Modeling | *Dr. Fan Zengyan* Jun 2026 – Present

- Investigating non-stationary asset dynamics and temporal volatility dependencies within high-frequency financial time-series datasets.
- Deriving stochastic predictive architectures to model underlying market mechanisms, bypassing the limitations of traditional stationary assumptions.
- Engineering data ingestion pipelines to process and structure multi-asset institutional streaming data from Bloomberg terminals for rigorous downstream quantitative estimation.

Stratified Non-Attendance Modeling in Clinical Datasets | *Dr. Fan Zengyan* Jun 2026 – Present

- Establishing a quantitative framework to isolate the demographic and behavioral covariates driving stratified patient non-attendance in localized clinical settings.
- Formulating survival analysis and categorical data models to evaluate time-to-event dynamics and map high-variance predictors of non-compliance.
- Deriving statistical diagnostics to adjust for real-world irregularities, applying heteroscedasticity-consistent estimators and non-parametric variance partitioning.

Causal Machine Learning for Treatment Heterogeneity Modeling | *Dr. Fan Zengyan* Oct 2025 – Jun 2026

- Investigated conditional treatment heterogeneity across high-dimensional feature spaces to derive welfare-maximizing assignment rules.
- Architected a tripartite causal machine learning framework, benchmarking Meta-Learners against Causal Forest Double Machine Learning to isolate individualized treatment effect surfaces.
- Enforced a rigorous identification protocol utilizing cross-validated propensity scores and deterministic boundary trimming to satisfy overlap assumptions and bound regularization bias.
- Synthesized localized individualized treatment effect vectors into an unsupervised clustering matrix to extract latent responder archetypes, validating results via an Optimal Policy Tree algorithm.

Multivariate Time Series Forecasting for Equity Price Prediction | *Dr. Zhou Tianyi* Jan 2025 – Nov 2025

- Evaluated the out-of-sample empirical stability of multi-horizon (1-, 5-, 20-day) return forecasting frameworks across NYSE-listed equities under non-linear asset dynamics.
- Conducted systematic comparative evaluations of classical linear estimators against sequential deep learning and contemporary self-attention architectures.
- Isolated economically meaningful technical signals via ensemble-based XGBoost feature selection pipelines, utilizing Recursive Feature Elimination and permutation importance.
- Validated the predictive superiority of attention mechanisms in capturing long-range dependencies through rigorous walk-forward backtesting regimes.

Data-Driven Framework for Enhancing Supply Chain Resilience | *Dr. Liu Ning* Nov 2024 – Aug 2025

- Investigated the theoretical and empirical performance boundaries of deep learning optimization architectures for stochastic inventory control against classical Predict-Then-Optimize paradigms.
- Synthesized dilated temporal Convolutional Neural Networks and Multi-Quantile Recurrent Neural Networks to model probabilistic, non-stationary demand distributions.
- Captured long-range temporal dependencies to optimize direct cost-functions in high-variance multi-SKU simulation environments.
- Demonstrated significant structural cost reductions and enhanced system resilience relative to traditional heuristic benchmarks.

Competency

Languages & Scripting: Python, R, SQL, LaTeX
Development & Tools: Git, VS Code, Visual Studio, Jupyter
Spoken Languages: English (Fluent), Mandarin (Native)